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Ionic liquid-mediated CO₂/N₂ separation in MoSe₂ nanochannels: a molecular dynamics perspective

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In recent years, supported ionic liquid membranes (SILMs) have attracted widespread attention due to their remarkable performance in gas separation. Two-dimensional (2D) nanomaterials, characterized by atomic-scale thickness, high specific surface area, and tunable mass transport pathways, are ideally suited for enhancing the separation efficiency of composite membranes through precise regulation of ionic liquids (ILs) and support architecture. In this study, a composite membrane system was constructed via molecular dynamics (MD) simulations, in which [BMIM][BF₄] was confined within molybdenum diselenide (MoSe₂) interlayers, and its performance in separating CO₂/N₂ mixtures was systematically evaluated. The separation efficiency was optimized by tuning key structural and operational parameters, including interlayer spacing, IL loading, and temperature. The results demonstrate that the optimal separation is achieved at 300 K with an IL loading ratio of 65% and an interlayer spacing of 4 nm. Furthermore, analysis of density distributions and cation orientation elucidated the microstructural characteristics and separation mechanisms of the membrane channels. The confined ILs within the MoSe₂ nanochannels exhibits a structure distinct from its bulk-phase counterpart, showing a stronger tendency to interact with gas molecules. The difference in gas solubility within the IL phase is identified as a key factor driving efficient separation. This study offers theoretical insights and design guidance for the development of 2D nanomaterial-based supported ionic liquid membranes (2D-SILMs) for CO₂/N₂ separation.

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1 Introduction

Since the Industrial Revolution, the extensive use of fossil fuels has substantially increased atmospheric carbon dioxide (CO₂) emissions, intensifying the greenhouse effect and making global warming one of the most pressing environmental challenges of the 21st century.^{1–4} In this context, CO₂ capture and separation technologies have emerged as critical pathways not only for climate change mitigation but also as foundational elements in establishing a low-carbon energy system. With continuous technological advances and cost reductions, their role has progressively shifted from a supplementary emission reduction option to a cornerstone technology for achieving carbon neutrality.⁵ Currently, the primary industrial CO₂ separation methods include chemical absorption, physical adsorption, cryogenic distillation, and membrane separation,

etc.^{6–8} Membrane separation technology operates on the principle of selective permeation, where gases pass through a membrane at different rates based on their physicochemical properties. This approach offers notable advantages such as low energy consumption (as it avoids phase changes), operational simplicity, and favorable environmental compatibility.^{9–11} Moreover, its capacity for continuous operation and scalability renders it both technically and economically viable for gas separation applications.^{12,13}

Ionic liquids (ILs) have attracted significant interest as novel functional materials for gas separation due to their distinctive physicochemical characteristics, which include tunable CO₂ affinity, negligible volatility, high thermal stability, and structural tunability.^{14–16} Unlike conventional volatile organic solvents, ILs offer two key advantages: (1) they remain liquid at or near room temperature, and (2) their cationic and anionic constituents can be purposefully tailored to modulate separation performance.^{17,18} Previous studies have demonstrated that ILs such as [BMIM][BF₄] exhibit high selectivity for CO₂/N₂ mixtures.^{19,20} However, the high viscosity of ILs often limits gas diffusion kinetics, while the restricted gas–liquid interface area can impede desorption efficiency.^{21,22} To overcome these challenges, supported ionic liquid membranes (SILMs) have been developed. In this configuration, ILs are confined within a porous support, which enhances structural integrity and

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increases the effective gas–liquid interfacial area, thereby improving mass transfer rates without compromising selectivity.^{23,24} This synergistic combination endows SILMs with considerable potential for efficient and scalable CO₂ separation.

In SILM technology, the choice of support material plays a vital role. Conventional polymeric membranes, such as polyimide and polysulfone, have been widely employed for CO₂/N₂ separation in flue gas from coal-fired power plants.^{25,26} However, their performance is fundamentally constrained by the well-known trade-off between permeability and selectivity, commonly referred to as the Robeson upper bound.²⁷ This inherent limitation significantly hampers the industrial implementation of membrane technology for efficient carbon capture. To address this challenge, growing attention has been directed toward novel two-dimensional (2D) materials.^{28,29} These materials offer unique advantages for gas separation, including atomic-level thickness, high specific surface area, and superior separation performance, which can facilitate rapid transport pathways for gas molecules. For instance, Fang *et al.*³⁰ developed a model based on functionalized 2D titanium carbide (Ti₃C₂) combined with ILs. Their findings indicated that functional groups indirectly influence CO₂ absorption and selectivity, whereas nanoslit width and additives markedly alter gas transport behavior and separation performance. In addition, Yan *et al.*³¹ employed all-atom molecular dynamics (MD) simulations to explore the separation mechanism of CO₂/CH₄ mixtures in graphene oxide (GO)-confined IL systems with varying interlayer spacings. They found that narrower interlayer spacing effectively enhances the solubility selectivity of CO₂/CH₄, while low-viscosity ILs contribute to improved diffusion selectivity. Among the systems studied, the [BMIM][TF₂N]/GO membrane exhibited the most favorable separation performance. Furthermore, Peng *et al.*³² fabricated a SILM by impregnating [BMIM][BF₄] onto MoS₂ substrates, which demonstrated outstanding CO₂ separation performance, with a selectivity of 131.42 and a permeance of 47.88 GPU. Wan *et al.*²⁰ combined MD simulations with density functional theory (DFT) calculations to investigate the separation mechanisms of three ILs ([BMIM][BF₄], [BMIM][PF₆], and [BMIM][Tf₂N]) within MoS₂-based SILMs. Their findings confirm the significant advantages of these systems in terms of CO₂ selectivity and the synergistic balance between permeation and blocking effects.

It is noteworthy that molybdenum diselenide (MoSe₂), a representative transition metal dichalcogenide (TMDC), shares structural and physicochemical similarities with MoS₂, yet exhibits distinct properties arising from its unique electronic and crystalline architecture. Specifically, selenium (Se) possesses a larger atomic radius, greater metallic character, and lower electronegativity than sulfur (S), leading to an expanded interlayer spacing, a narrower bandgap, and enhanced out-of-plane polarizability in MoSe₂.^{33,34} These attributes collectively result in a surface with lower polarity and reduced surface energy, which render MoSe₂ highly promising for applications spanning electronic and optoelectronic devices, energy storage and conversion systems, catalytic processes, and membrane separation.^{33,35} The structural parallels and differences among

such materials offer a wide design space for developing next-generation high-performance separation membranes. Against this background, MoSe₂ was chosen as the support material in this work. Using MD simulations, a sandwich-structured [BMIM][BF₄]@MoSe₂ composite membrane was constructed by confining [BMIM][BF₄] between two MoSe₂ nanosheets to achieve efficient CO₂/N₂ separation. The separation performance of the composite membrane was optimized by tuning key parameters such as interlayer spacing, ILs loading, and temperature. Moreover, the following microscopic characteristics were investigated in detail: number density distributions of the ILs and gas molecules, 2D density profiles, orientation distributions of cations within the confined channel, inter-component interaction energies, radial distribution functions (RDFs), free volume properties, and gas-diffusion dynamics. Through these analyses, this study elucidates the microstructural features of the [BMIM][BF₄]@MoSe₂ composite membrane, the dynamic behavior of ILs and gas molecules, and the interaction mechanisms among the constituent components. The results provide important theoretical insights and design guidelines for the application of 2D-SILMs in CO₂/N₂ separation.

2 Models and methods

In the simulation system, a MoSe₂ membrane measuring 4.66 nm × 4.61 nm was employed. To ensure adequate diffusion space for gas molecules, the simulation box length along the z-axis was set to 20.61 nm. Nanochannels with interlayer spacings of 3 nm, 4 nm, and 5 nm were constructed to investigate nano-confinement effects. The corresponding numbers of IL ion pairs placed in the central membrane region were 132, 176, and 220, respectively, while maintaining a constant volume loading ratio of 65%. Fig. 1a illustrates a representative system configuration with a 4 nm interlayer spacing, and the molecular structures of [BMIM]⁺ and [BF₄][−] are shown separately in Fig. 1b and c. The ILs were randomly packed into the nanochannels using the PACKMOL program.³⁶ Following common simulation practices,^{20,30,37,38} an equimolar (1 : 1) CO₂/N₂ mixture (100 molecules of each) served as the feed gas. Simulations with a realistic flue-gas composition (15 vol% CO₂, Fig. S1) confirmed that the membrane maintains significant selectivity, demonstrating its robust performance across varying feed conditions. The left region served as the mixed-gas feed chamber, whereas the right side was kept under vacuum. Periodic boundary conditions were applied in all three directions. During the

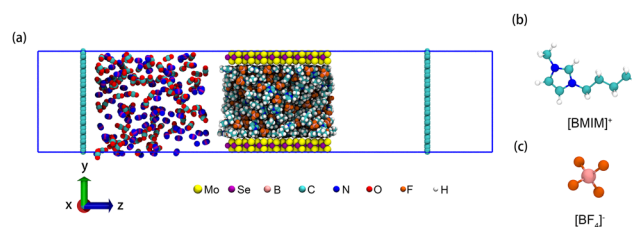


Fig. 1 (a) Schematic of the simulation system with a 4 nm interlayer spacing. Molecular structure of (b) [BMIM]⁺ and (c) [BF₄][−].

simulations, all MoSe₂ and graphene sheets were held fixed^{38,39}—a conventional and well-justified practice in nanoconfined fluid simulations—in order to isolate the key effects of interlayer spacing and IL structuring on gas transport. This approach is physically supported by the high in-plane stiffness of these materials, which ensures that thermal fluctuations do not qualitatively alter the geometric confinement or the resulting size-sieving mechanism. Based on the ideal gas equation of state, the initial feed-side pressure for the system with a 4 nm interlayer spacing was set to 7.8 MPa. This value lies within the conventional 1–10 MPa range commonly used in MD simulations^{20,37,38} to achieve practical sampling efficiency. Importantly, a control simulation at a reduced pressure of 3.9 MPa verified that the membrane retains considerable CO₂/N₂ selectivity (Fig. S1). These results indicate that the observed separation capability is intrinsic and robust across the studied pressure range.

All MD simulations were performed using the GROMACS 5.1.5 package.⁴⁰ The OPLS-AA force field^{41,42} was adopted, which has been validated in prior studies of graphene,^{41,42} MoSe₂,⁴³ CO₂,⁴⁴ N₂,⁴⁴ and [BMIM][BF₄].⁴⁵ CO₂ and N₂ molecules were represented using a three-site model. In the graphene sheets, the atomic charges of C atoms were set to zero, whereas those of Mo and Se atoms were assigned as +0.77e and −0.385e, respectively.^{46,47} The fixed-charge model—a widely employed approach in classical MD for 2D material/fluid systems^{19,47,48}—adequately captures surface electrostatics and enables simulations with large size and time-scale, as performed in this study. Although such charge model may neglect polarization and moderately attenuate CO₂-ion interactions,

the effect is insignificant⁴⁹ and consistent across all the studied systems. Moreover, given N₂'s significantly lower polarizability than CO₂, considering polarization (*e.g.* employing polarizable force fields^{50,51}) would further enhance their adsorption and transport differences, thus increasing the CO₂/N₂ selectivity. The system temperature was maintained at 300 K by the Nosé–Hoover thermostat,^{52,53} with a coupling constant of 0.5 ps. van der Waals interactions were truncated at 1.2 nm, and long-range electrostatic interactions were treated using the particle mesh Ewald (PME) method,⁵⁴ with a cutoff distance of 1.2 nm. Bonds involving hydrogen atoms were constrained with the LINCS algorithm.⁵⁵ The simulation protocol comprised three stages: (1) energy minimization of the initial configuration to eliminate atomic overlaps; (2) a 2 ns NVT equilibrium using an annealing protocol (300 K → 800 K → 300 K) to facilitate system relaxation; (3) a final 200 ns production run, with the final 50 ns of trajectory data used for statistical analysis. All results were averaged over five independent simulation replicates.

3 Results and discussion

3.1 Separation performance

3.1.1 Effect of interlayer spacing. Interlayer spacing is a key structural parameter governing membrane separation performance. This study examined three different spacings—3 nm, 4 nm, and 5 nm—with corresponding IL loadings of 132, 176 and 220 ion pairs, respectively, while maintaining a constant volume loading rate of 65%. By analyzing gas absorption

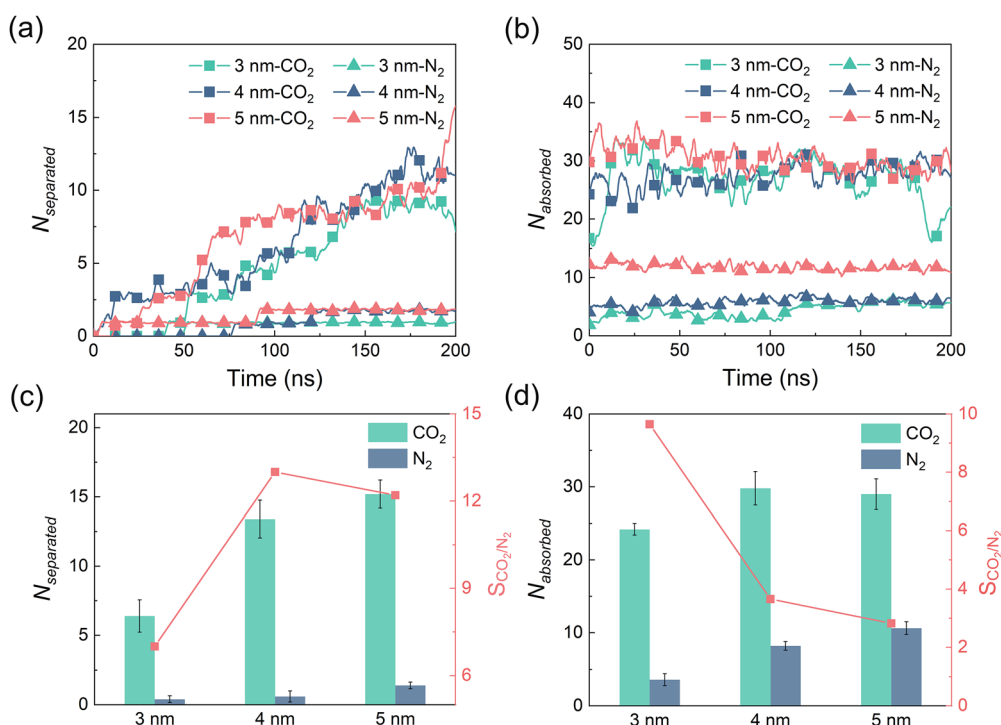


Fig. 2 For different interlayer spacings: (a) gas permeation count and (b) gas absorption capacity as a function of time; (c) gas permeation count versus separation selectivity; (d) absorption capacity and absorption selectivity.

characteristics (including absorption capacity and absorption selectivity) within the channels at each spacing and comparing them with actual separation performance metrics (such as permeation flux and separation selectivity), the structure–performance relationship between interlayer spacing and membrane efficiency was elucidated. As shown in Fig. 2a, during the separation process, the permeation count of CO₂ exhibits slight upward fluctuations, whereas that of N₂ remains relatively stable. Notably, the absorption curve of CO₂ molecules displays considerable variation in Fig. 2b, which can be attributed to the dynamic absorption–desorption equilibrium at the gas–liquid interface. A comparative analysis of Fig. 2c and d reveals that as the interlayer spacing increases from 3 nm to 5 nm, the gas permeation flux increases substantially. In contrast, the trend in absorption capacity differs. Overall evaluation indicates that the separation performance is optimal at a 4 nm spacing, achieving the highest separation selectivity (up to 13) while also exhibiting higher CO₂ absorption capacity than the 5 nm case. It is worth noting that at the 3 nm spacing, absorption selectivity is the highest (9.65), whereas separation selectivity is the lowest (7). This result stems from the pronounced nanoconfinement and enhanced interfacial interactions at reduced spacings, which promote a transition of the ILs into a predominantly surface-adsorbed state. While this state improves the capacity of the ILs to selectively absorb certain gases, it also severely restricts molecular mobility, thereby hindering gas diffusion and reducing diffusion selectivity. The interplay of these competing effects leads to the lowest overall membrane separation at this scale.

Furthermore, the permeance is defined as the permeability (P) divided by the membrane thickness (l), i.e., P/l , and is calculated as follows:

$$\frac{P}{l} = \frac{n}{A \cdot \Delta p \cdot t}$$

where n is the molar quantity of gas molecules (mol), A denotes the cross-sectional area of the membrane channel (m²), Δp indicates the transmembrane pressure difference (Pa), t is the duration of gas separation (s), and l is the membrane thickness. By definition, 1 GPU = 3.35×10^{-10} mol m⁻² s⁻¹ Pa⁻¹. The calculated permeance results show that at a layer spacing of 3 nm, the CO₂ permeance reaches 1.92×10^3 GPU, whereas that of N₂ is 5.96×10 GPU. When the spacing increases to 4 nm, the CO₂ permeance rises significantly to 2.67×10^3 GPU, and the N₂ permeance reaches 8.95×10 GPU. At a spacing of 5 nm, the CO₂ permeance is 2.68×10^3 GPU and the N₂ permeance is 2.09×10^2 GPU. These results indicate that an optimal interlayer spacing (4 nm) can strike the best balance between selectivity and permeance, thereby providing a valuable theoretical guideline for the design of high-performance gas separation membranes.

3.1.2 Effect of IL volume loading ratio. Taking an interlayer spacing of 4 nm (corresponding to 176 IL ion pairs) as the reference, the influence of the IL volume loading ratio on separation performance was meticulously examined by adjusting the number of ion pairs to 160 and 192, corresponding to

loadings of 59% and 70%, respectively. As depicted in Fig. 3, reducing the loading to 59% increases the permeance of both CO₂ and N₂, but lowers the separation selectivity to 10.1. This suggests that selective transport channels formed under insufficient IL loading cannot effectively restrict N₂ permeation, which compromises separation performance. Conversely, when the loading is increased to 70%, the overall gas permeance decreases, with CO₂ permeance dropping notably, leading to a separation selectivity of 9.6. This indicates that excessive IL loading, while almost completely suppressing N₂ permeation, significantly raises mass transfer resistance and severely hinders CO₂ transport. A similar trend is observed at a 5 nm interlayer spacing (Fig. S2), confirming that IL loading non-monotonically governs separation efficiency. This behavior arises from the nanoconfined filling state of the IL: insufficient loading leads to discontinuous selective channels, whereas excessive loading causes channel blockage or elevated viscosity, hindering diffusion. The mechanism—and the resulting performance optimum—shows little dependence on interlayer spacing, indicating that an optimal IL loading is intrinsic to the IL-channel system. Here, 65% IL volume loading yields the best balance between selectivity and permeance, consistent with prior studies which also identify an intermediate IL loading as key to optimizing the performance of 2D-based composite membranes.^{19,39}

3.1.3 Temperature effect. To further optimize separation conditions, the performance of the CO₂/N₂ separation system was evaluated across a temperature range of 300–350 K, using a fixed interlayer spacing of 4 nm and an IL volume loading of 65%, as illustrated in Fig. 4. The results show that as temperature increases from 300 K to 350 K, the permeation flux of CO₂ rises significantly, while its absorption capacity within the channel decreases. This trend can be attributed to enhanced CO₂ desorption from the IL at higher temperatures, which subsequently promotes the diffusion of desorbed CO₂ molecules through the available free volume in the membrane channels. The overall behavior thus reflects a coupled effect of increased permeation and reduced absorption. In contrast, although the permeation flux of N₂ also increases with temperature, its absorption capacity within the channel remains

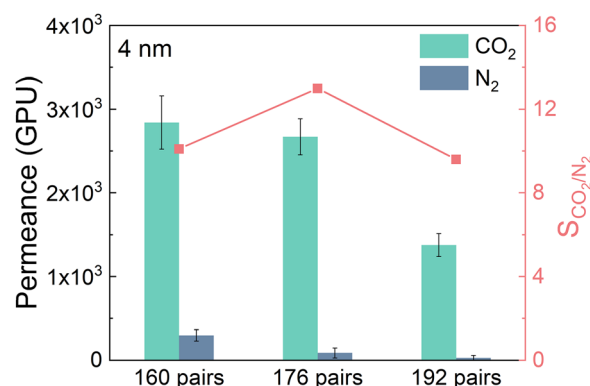


Fig. 3 Gas permeance and separation selectivity versus IL pairs at an interlayer spacing of 4 nm.

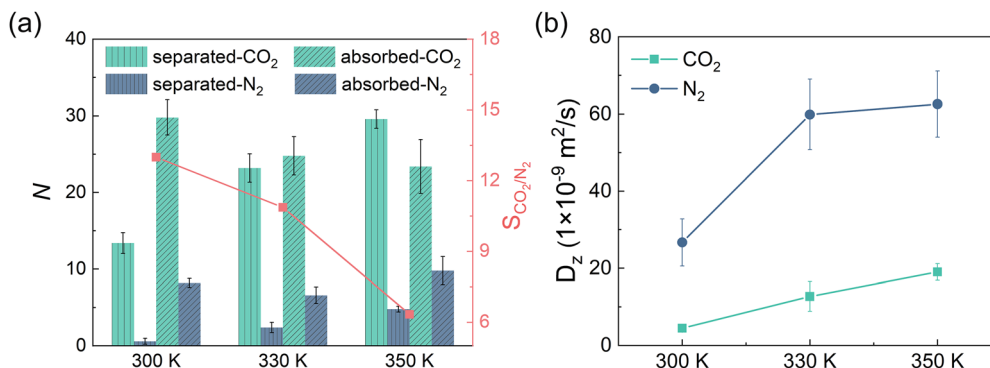


Fig. 4 (a) Effect of temperature on gas permeation performance; (b) self-diffusion coefficients of gases as a function of temperature.

largely unchanged. This stability stems from the inherently low solubility of N_2 in ILs, where transport is governed predominantly by a free-volume diffusion mechanism, rendering its absorption behavior less sensitive to temperature variation. Notably, as temperature rises, the separation selectivity of the composite membrane for CO_2/N_2 decreases from 13 to 6.34, indicating a pronounced temperature-dependent regulation of gas separation performance. This finding provides a theoretical basis for designing and optimizing temperature-responsive gas separation membranes. From the perspective of molecular diffusion theory, the influence of temperature on separation performance can be further elucidated. As shown in Fig. 4b, an increase in temperature intensifies the thermal motion of gas molecules, leading to a marked enhancement in their diffusion coefficients—a behavior consistent with the Arrhenius-type diffusion relationship.^{56,57} Owing to the smaller kinetic diameter of N_2 and its weaker intermolecular interactions, its self-diffusion coefficient remains higher than that of CO_2 across all temperatures studied. As temperature rises, the diffusion capacity of N_2 increases more substantially than that of CO_2 , resulting in a greater relative enhancement in N_2 permeation flux. This differential enhancement directly manifests as a decline in overall separation selectivity.

The permeance of CO_2 and N_2 was systematically evaluated at 300 K, 330 K, and 350 K. The corresponding Arrhenius plots of $\ln(P/l)$ versus $1000/T$ (Fig. S3) show a highly linear relationship, confirming that gas transport in this system follows Arrhenius

behavior over this temperature range. Linear fitting yielded apparent activation energies of $E_{a,CO_2} = 13.68 \pm 3.66 \text{ kJ mol}^{-1}$ and $E_{a,N_2} = 32.98 \pm 7.06 \text{ kJ mol}^{-1}$. These values correspond to the energy barriers associated with gas diffusion through the nano-confined ILs, which involve processes such as disrupting local ion pairs or navigating transient free-volume pathways. The significantly higher activation energy for N_2 results in a stronger temperature dependence of its permeance, which in turn explains the observed decrease in CO_2/N_2 selectivity with increasing temperature (Fig. 4a). To further examine the influence of free volume on ion transport, we also computed the free volume fraction within the confined channels across the same temperature range (Fig. S4a). The results show that between 300 K and 350 K, the free volume fraction varies by less than 0.02%, which can be considered essentially constant. This indicates that no significant thermal expansion occurs in the confined ILs over this temperature interval.

3.2 IL density distribution

To elucidate the internal structural characteristics of the nano-channels, Fig. 5 compares the number density profiles of $[BMIM]^+$ and $[BF_4]^-$ along the y-axis under three different interlayer spacings. To ensure the reliability of these profiles, the distributions were computed with high-resolution grids using the gmx density module in GROMACS. Moreover, the nearly complete overlap between the density profiles obtained from two consecutive 25-ns sub-intervals (150–175 ns and

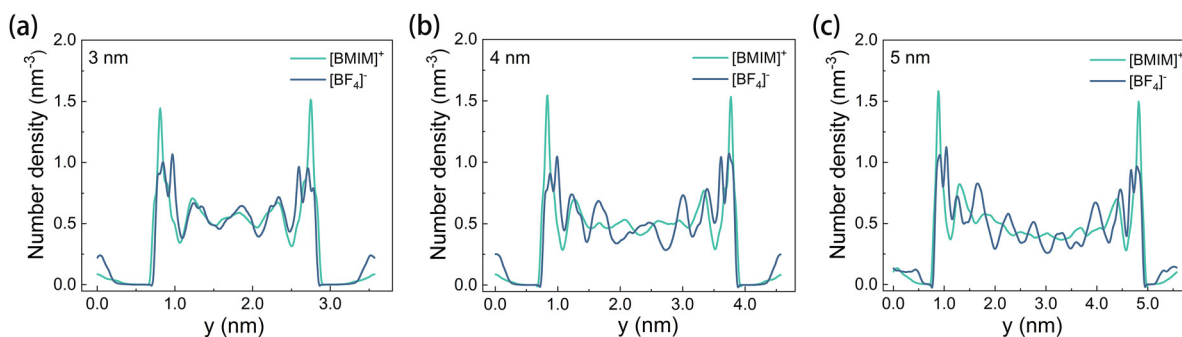


Fig. 5 Number density distributions of $[BMIM]^+$ and $[BF_4]^-$ along the y-axis for different interlayer spacings: (a) 3 nm; (b) 4 nm; (c) 5 nm.

175–200 ns) confirms the structural convergence of the system (Fig. S5). The results indicate that cations exhibit significantly higher enrichment in the interfacial regions than anions, as evidenced by the markedly stronger density peaks of cations on both sides of the interface. This confirms the predominance of cation–surface interactions. At an interlayer spacing of 3 nm, the ILs adopt a distinct five-layer distribution: two high-density adsorption layers form near the interfaces, while three lower-density layers occupy the central region. The ion density within the confined interfacial layers is substantially higher than that in the intermediate layers. The overlapping peak positions of cations and anions indicate that strong interionic interactions are maintained despite the interfacial confinement. As the interlayer spacing increases, the ILs continue to display a stratified structure. Notably, when the spacing is increased from 4 nm to 5 nm, the density of the confined IL layers remains largely unchanged, suggesting structural stability within this range. However, unlike the 3 nm case, spatial separation of ions emerges—specifically, cation density peaks disappear in the central layer, and ion pairing is preserved only within the confined adsorption layers. These observations demonstrate that the MoSe₂ surface strongly influences the distribution and structural organization of the ILs. This effect is more pronounced at smaller interlayer spacings, leading to a more ordered arrangement that facilitates the formation of gas-diffusion pathways. At the 3 nm spacing, the ILs exist entirely in an adsorbed state, as reflected by the five-peak density profile. In contrast, only a double-peak adsorption structure forms at a single interface, with no bulk-like region present. Under such conditions, gas transport is primarily governed by an absorption-mediated mechanism.

Moreover, the distinct surface properties of MoSe₂ induce unique interactions with ILs, yielding a more pronounced layered structure of confined ILs compared to systems based on MXene,³⁰ GO,³¹ or MoS₂,²⁰ suggesting that MoSe₂ exerts stronger spatial confinement and structural regulation on ILs. Fig. S6 further illustrates the temperature-dependent evolution of the layered IL structure along the channel's y-axis. As temperature increases, the ion density within the central IL layer decreases, whereas the confined layer adjacent to the MoSe₂ surface is slightly enhanced. This suggests that thermal energy loosens the ion arrangement in the channel center, increasing structural disorder and thereby facilitating gas diffusion, while simultaneously stabilizing the interfacial ionic layer.

Fig. 6 illustrates the 2D density distribution of [BMIM]⁺ and [BF₄][−] ions in channels with three different interlayer spacings. Regardless of spacing, cations consistently exhibit obvious interfacial enrichment, forming a distinct bilayer structure within the confined IL system, while anions remain more broadly dispersed. These observations are consistent with the one-dimensional number density analysis. At a spacing of 3 nm, ions in the central region are sparsely distributed, and free volume regions are randomly dispersed. The consequent lack of well-defined transport pathways leads to limited gas permeance. As the interlayer spacing increases, the ion distribution in the central region gradually becomes sparse. Notably, at 5 nm, cations and anions adopt a characteristic hourglass-shaped distribution. This structural evolution concentrates free volume predominantly in the channel center, effectively reducing diffusion resistance and significantly enhancing gas permeance. Furthermore, when combined with Fig. 5, it is evident that increasing the interlayer spacing primarily thickens the

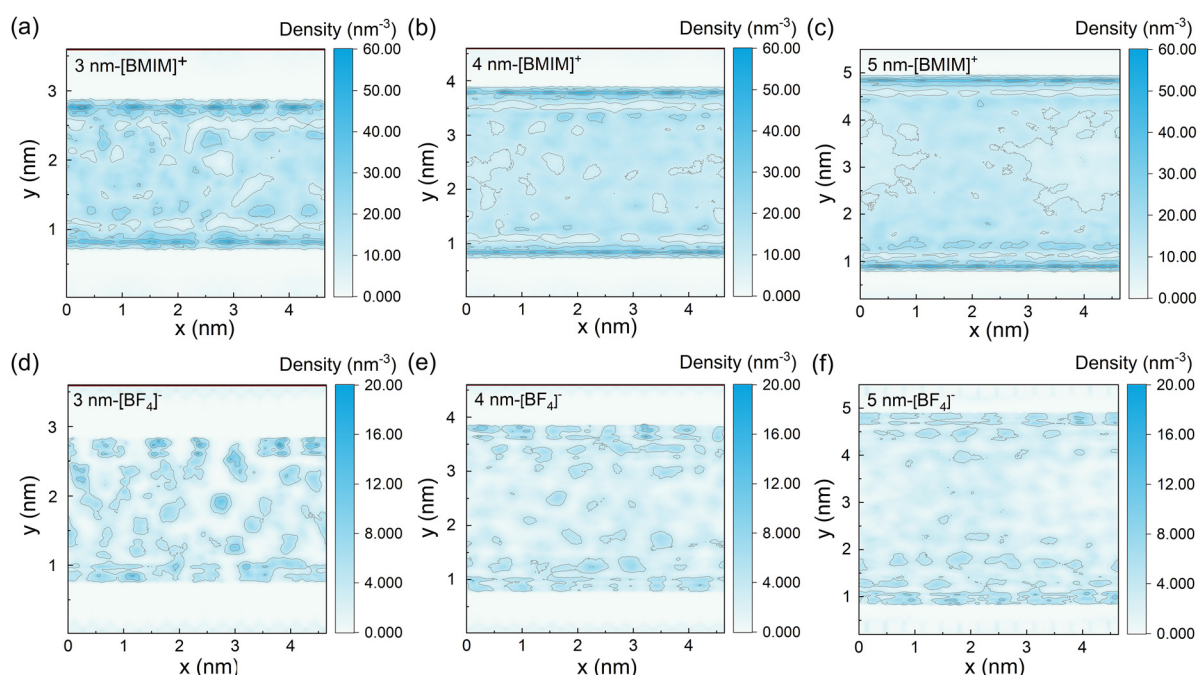


Fig. 6 2D density distributions of (a)–(c) [BMIM]⁺ and (d)–(f) [BF₄][−] on planes perpendicular to the z-axis for interlayer spacings of (a) and (d) 3 nm, (b) and (e) 4 nm, (c) and (f) 5 nm.

intermediate IL layer, while the thickness of the confined interfacial IL layers remains largely unchanged. Since CO_2 is mainly adsorbed within the confined IL layers near the two nanochannel surfaces, and the intermediate IL layer exhibits relatively low gas absorption capacity, this accounts for the higher CO_2 absorption observed at a 4 nm spacing compared to 5 nm, as shown in Fig. 2d.

Given the limited structural relevance of the spherically symmetric $[\text{BF}_4]^-$ anions, this study focuses on the orientational ordering of the $[\text{BMIM}]^+$ cation to further elucidate the structural characteristics of the confined ILs. As depicted in Fig. 7a, vector A is defined along the line connecting the two N atoms of the imidazole ring, vector B connects the ring-junction N atom to the terminal C atom of the alkyl chain, and the vector Z is normal to the MoSe_2 surface. The spatial orientation of cations is quantified by the angle θ between A and Z , and the angle φ between B and Z . As shown in Fig. 7b, under gas-free conditions, both θ and φ exhibit pronounced peaks near 90° , indicating that cations preferentially align parallel to the MoSe_2 surface. This orientation preference stems from strong cation-surface interactions, which are maximized in the parallel configuration and simultaneously facilitates the formation of gas transport pathways. Following gas-separation runs, the orientation distribution becomes notably more random, and the initial peak at 90° dissipates. This confirms a distinct interaction between gas molecules and the ILs, which—though

weaker than the surface-IL interaction—is sufficient to disrupt the ordered cationic arrangement. Furthermore, Fig. 7c and d present the orientation distributions of the imidazole ring and alkyl chain at different interlayer spacings. Across all three spacings, both the ring and alkyl chain maintain a preferred parallel orientation relative to the MoSe_2 surface (characteristic angle $\sim 90^\circ$). This consistency arises from the strong spatial confinement imposed by the MoSe_2 surface, implying that the composite membrane retains separation capability regardless of the interlayer spacing. Notably, as the spacing increases, the angular distribution of the alkyl chain broadens slightly, though the extent of this change remains limited.³¹ This indicates that larger spacings primarily increase the thickness of the intermediate IL layer,⁵⁸ where a greater proportion of ions experience weaker surface constraints, leading to a modest expansion in orientational freedom.

3.3 Gas distribution

To further elucidate the gas separation process, Fig. 8a and b shows the number density profiles of CO_2 and N_2 along the y - and z -axis for an interlayer spacing of 4 nm, respectively. The results reveal that CO_2 exhibits a distinct high-density peak near the MoSe_2 surface, with its peak position closely aligned with those of the cations and anions. This confirms the high solubility of CO_2 in the IL and indicates that its distribution is strongly influenced by the spatial arrangement of the IL.

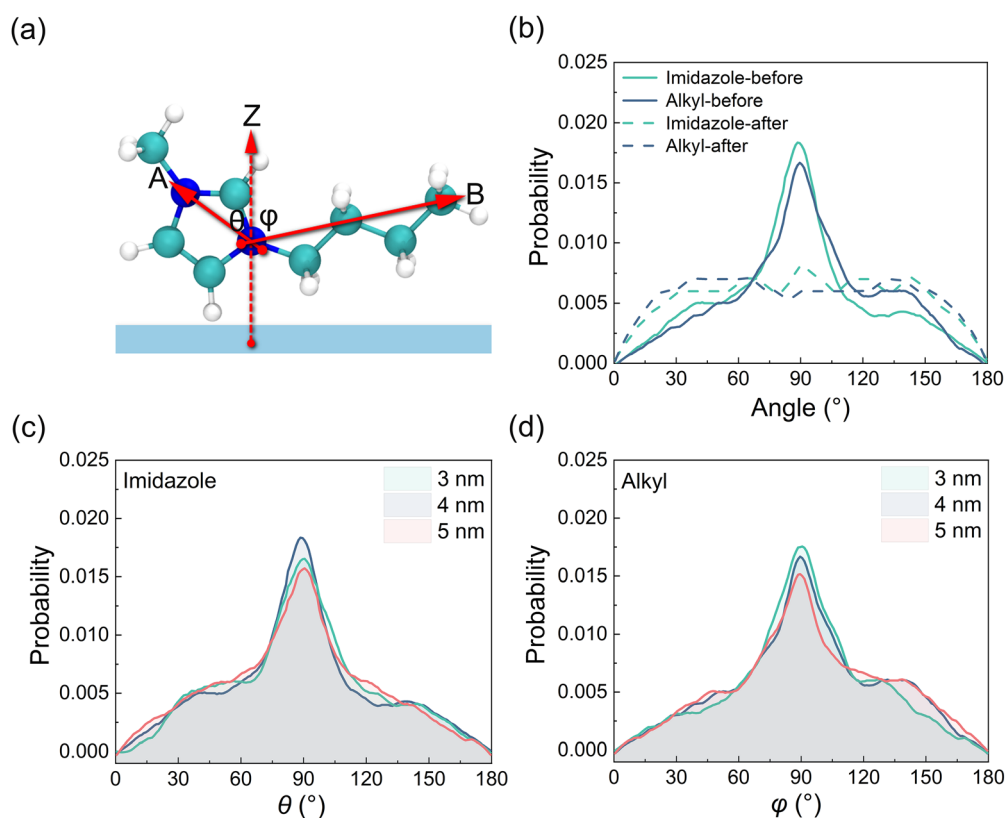


Fig. 7 (a) Schematic of the orientation vectors for the imidazole ring and alkyl chain; (b) angular distributions of the imidazole ring and alkyl chain before gas introduction and after gas separation; (c) angular distribution of the imidazole ring and (d) alkyl chain prior to gas introduction at three different interlayer spacings.

Crucially, the enriched interfacial CO_2 layer acts not as a static barrier but as a dynamic precursor for efficient long-range transport. The linear increase in permeated CO_2 over time (Fig. 2a) confirms a steady transmembrane process, which would be impeded if inlet enrichment caused blockage. Although a sharp density peak exists at the inlet (Fig. 8b), the number of CO_2 molecules inside the channel fluctuates around a stable average (Fig. 2b). Moreover, CO_2 molecules within this region retain significant surface mobility even under strong anion interactions. In contrast, N_2 shows weak interactions with both the material surface and the IL, resulting in uniform distribution along the y -axis without a pronounced density peak (Fig. 8a). The 2D density profiles (Fig. 8c and d) reveal that N_2 penetration is limited; entering molecules are confined to free-volume regions near the entrance and do not effectively dissolve into the IL phase, as also reflected in the declining number density along the channel front. Comparison with IL distribution (Fig. S7b) further supports this picture. Thus, selective separation arises from the distinct solubility-driven behaviors: CO_2 forms a dynamic, transport-active enriched interface, whereas N_2 remains poorly dissolved and spatially restricted. This interpretation aligns with the reported higher Henry's constant for N_2 in comparable systems.⁵⁹

3.4 Interaction energy

To clarify the microscopic mechanism of gas separation, the interaction energies between gas molecules and the composite

membrane were evaluated. As shown in Fig. 9a, the interaction energy between N_2 and MoSe_2 is negligible (close to zero), while that between N_2 and the ILs, though slightly higher, remains significantly weaker than the CO_2 - MoSe_2 interaction. The fluctuation of interaction energies around their respective means during the final 50 ns of simulation suggests that the system reached a stable state. Among the four pairs analyzed, the CO_2 -ILs interaction is the strongest. Notably, the CO_2 -ILs energy exhibits considerable fluctuations, which is attributed to the high solubility of CO_2 in the ILs and facilitates effective permeation of CO_2 through the membrane. Conversely, due to the low solubility of N_2 in the ILs, N_2 molecules are predominantly adsorbed near the channel entrance and cannot penetrate the membrane structure; hence, N_2 -composite membrane interaction remains relatively stable with only minor fluctuations. Throughout the separation process, the interaction strength of both CO_2 and N_2 with the ILs markedly exceeds that with MoSe_2 , underscoring the dominant role of the ILs in the separation performance of the composite membrane. Fig. 9b presents a comparative analysis of the average interaction energies at three interlayer spacings. The results show that the cation- MoSe_2 interaction energy consistently exceeds the anion- MoSe_2 energy, with the former displaying characteristic attraction and the latter repulsion. However, both ions-surface interactions are much weaker than the cation-anion interaction within the IL. As the interlayer spacing increases, the cation-anion interaction strengthens, whereas the surface-

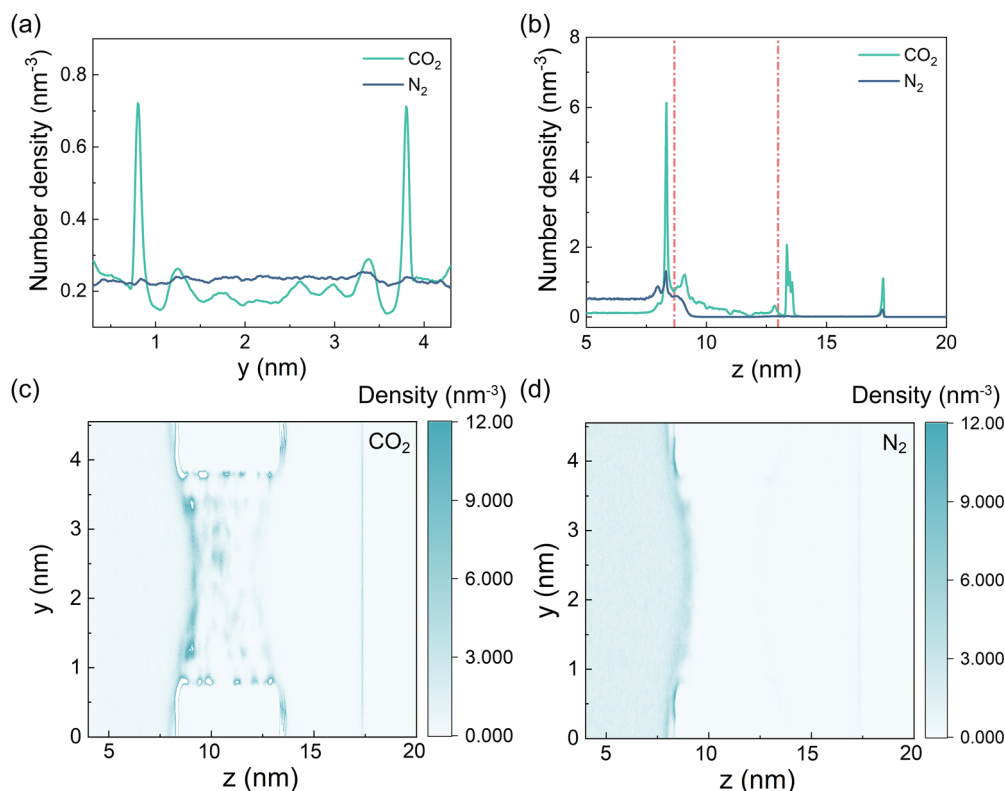


Fig. 8 At an interlayer spacing of 4 nm: number density distribution of gas molecules along the (a) y -axis and (b) z -axis; 2D density distribution of (c) CO_2 and (d) N_2 .

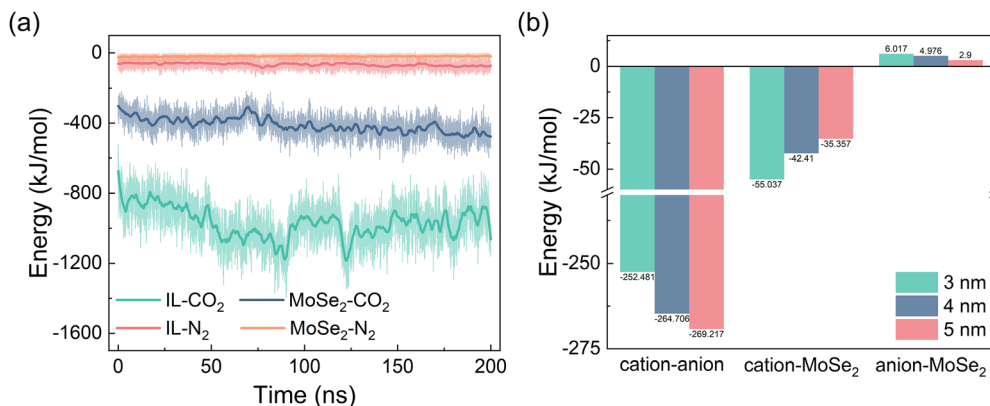


Fig. 9 (a) Interaction energies between gas molecules and the ILs/MoSe₂ at a 4 nm interlayer spacing; (b) comparison of average interaction energies between cations–anions and ions–MoSe₂ at different interlayer spacings.

ion interaction weakens considerably, indicating that a larger interlayer spacing significantly reduces the surface-confinement effect on ions. It should be emphasized that the strength of anion–cation interactions in ILs directly influences their free volume characteristics.^{60,61} When interionic interactions are weak, the system free volume increases, which favors interactions with CO₂ molecules. Accordingly, Fig. S4b quantitatively evaluates the free volume fractions under different spacings. At a constant IL volume loading of 65%, the free volume fraction within the channel decreases as the spacing increases from 3 nm to 5 nm.

3.5 Radial distribution function

To elucidate the interaction mechanism of CO₂ and N₂ within the IL, RDFs for cation–CO₂, anion–CO₂, cation–N₂, and anion–N₂ were analyzed at three interlayer spacings (Fig. 10). Representative atoms were selected: the central C atom of the cation, the B atom of the anion, the C atom of CO₂, and the N atom of N₂. The results show that the cation–N₂ RDF lacks distinct peaks, while the anion–N₂ RDF exhibits only a minor peak (peak height <1), confirming the weak interaction between N₂ and the ILs, which aligns with the energy analysis presented in Fig. 9. In contrast, the RDFs for both cation–CO₂ and anion–CO₂ pairs display bimodal profiles, albeit with distinct structural origins. The anion–CO₂ RDF features a sharp and intense

first peak at ~0.38 nm—significantly higher than the second peak—which reflects strong electrostatic interactions between the CO₂ quadrupole and the anions. Conversely, the two peaks in the cation–CO₂ RDF are of comparable intensity and closely spaced, a feature attributed to the spatial confinement of CO₂ molecules within the constrained IL layer due to steric hindrance from the cations. This interpretation is further supported by the density distribution analyses presented in Fig. 5 and 6. Moreover, spatial distribution snapshots of CO₂ within the channel (Fig. S8) reveal that in regions of strong confinement near the surfaces, CO₂ molecules adopt a flattened orientation parallel to the interface to maximize intermolecular interactions, while toward the channel center, where confinement weakens, their orientation becomes more vertical or disordered. This visualizes how combined spatial and electrostatic constraints govern molecular arrangement. As the interlayer spacing increases, the characteristic RDF peaks attenuate and the orientational distribution broadens. Such changes align with the observed reduction in interaction energy at larger spacings (Fig. S9), thereby providing a microscopic perspective on how nanoconfinement modulates gas adsorption behavior.

3.6 Mean square displacement and self-diffusion coefficient

To further investigate gas-transport dynamics, the MSD of CO₂ and N₂ in channels with different spacings was calculated.

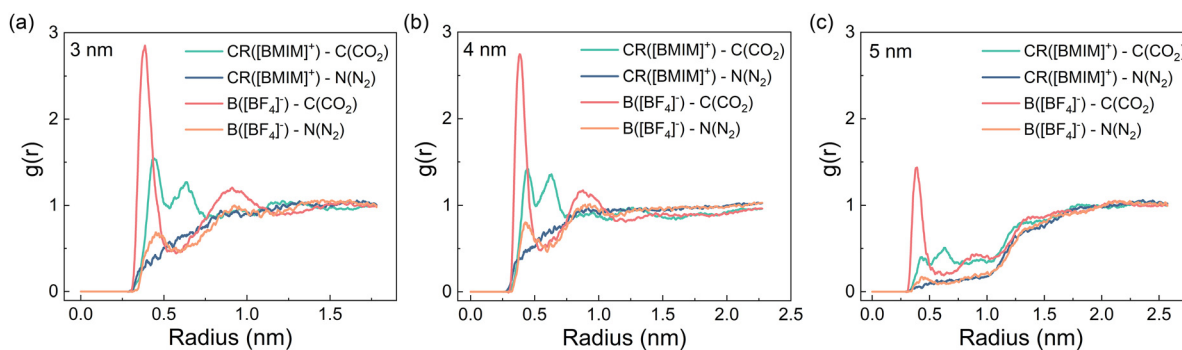


Fig. 10 RDFs of cation–gas and anion–gas systems at different interlayer spacings: (a) 3 nm, (b) 4 nm, (c) 5 nm.

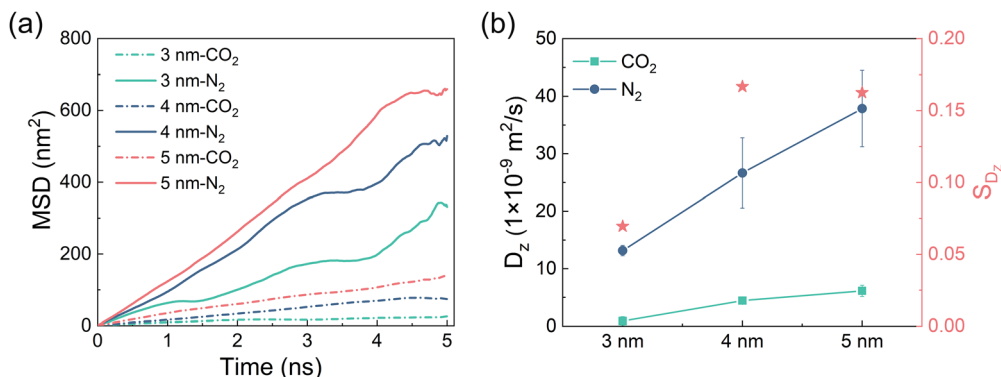


Fig. 11 Transport properties of gas molecules within nanochannels at different interlayer spacings: (a) MSD; (b) self-diffusion coefficients.

Based on Einstein's equation,⁶² the self-diffusion coefficient of each gas was obtained from the linear regression of MSD *versus* time. Given the spatial confinement effect of MoSe₂ along the y-axis on gas diffusion, this analysis primarily focuses on the self-diffusion coefficient along the z-axis. As shown in Fig. 11, under all conditions tested, both the MSD and self-diffusion coefficient of N₂ are significantly higher than those of CO₂. Specifically, the self-diffusion coefficients of CO₂ at three spacings are 0.9168×10^{-9} , 4.446×10^{-9} , and $6.156 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$, respectively, while those of N₂ are 13.17×10^{-9} , 26.66×10^{-9} , and $37.86 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$, respectively. The resulting CO₂/N₂ diffusion selectivities at all three spacings are below 1, with the value closest to 1 at the 4 nm spacing. Diffusion selectivities below 1 for CO₂/N₂ have been previously reported in similar 2D-material/IL composite membranes.^{58,63} This phenomenon arises from the distinct thermodynamic and kinetic behaviors of the gases in the IL phase. CO₂ exhibits strong affinity and high solubility, evidenced by its substantial interaction energy with IL ions (Fig. 9a) and pronounced density enrichment near IL layers (Fig. 8a). These interactions prolong its residence time and restrict its translational mobility. In contrast, N₂ shows negligible interaction energy and no significant density accumulation, allowing it to diffuse more rapidly through available free-volume pathways.

Gas permeability follows the solution-diffusion model, $P = D \times S$,⁶⁴ where D is the diffusion coefficient and S is the solubility coefficient. Here, the membrane thickness is defined as the distance between the two Gibbs dividing surfaces^{65,66} that separate the IL phase from the bulk vapour phases, as determined from the number density profiles along the z-axis (Fig. S7a). For the 4 nm interlayer spacing, the CO₂/N₂ permeation selectivity is 29.84. Although the diffusion selectivity is only 0.167, it is fully compensated by a very high solubility selectivity of 178.51 (Table S1). Thus, while CO₂ diffuses more slowly, its exceptional solubility establishes a steep concentration gradient and a strong permeation driving force. Conversely, N₂ possesses negligible solubility and thus minimal net flux, despite its higher mobility. Therefore, separation in the MoSe₂/IL nanochannel is governed primarily by thermodynamic dissolution selectivity rather than kinetic diffusion sieving. The dominant role of solubility not only offsets the

unfavorable diffusion but fundamentally underpins the high overall selectivity, confirming that transport in this composite membrane follows a solution-diffusion mechanism in which solubility is the key performance determinant.

4 Conclusions

In this study, a sandwich-structured composite membrane consisting of [BMIM][BF₄] and MoSe₂ was systematically investigated *via* MD simulations for its CO₂/N₂ separation performance. The results indicate that an optimal interlayer spacing of 4 nm and a volume loading of 65% yields the best separation efficiency. Increasing temperature enhances the permeation flux of both CO₂ and N₂ but simultaneously lowers the overall separation selectivity. Mechanism analysis reveals that strong interactions of CO₂ with both the MoSe₂ surface and the IL promote significant enrichment of CO₂ at the interface and within the IL layer. As the interlayer spacing widens, the interaction between CO₂ or MoSe₂ and the [BMIM]⁺/[BF₄][−] ions weakens, while cation–anion interactions become more dominant. Importantly, gas transport in this system follows a solution-diffusion mechanism. Although the self-diffusion coefficient of N₂ exceeds that of CO₂ at all interlayer spacings examined, the markedly higher solubility of CO₂ in the IL is the primary determinant of selectivity, enabling effective CO₂/N₂ separation. This study offers a molecular-level understanding of gas separation mechanisms in 2D-SILMs, providing theoretical guidance for the design of high-performance gas-separation membrane materials, particularly in optimizing membrane architecture and operating parameters. However, the simulation methodology—specifically, establishing a chemical potential gradient using a downstream vacuum region—may introduce artifacts. These include (1) an artificially high driving force, potentially overestimating permeation rates, and (2) structural deviations of ILs at the simulated membrane–vacuum interface from a real liquid–gas interface. Moreover, simulated permeance values ($\sim 10^3$ GPU) are higher than typical experimental data for supported IL membranes,^{32,67} a discrepancy may be attributed to the idealized defect-free structure modeled, the neglect of capillary and

viscous flow effects, and simplifications of the simulated membrane composition. While the simulations clarify molecular-level transport and intrinsic material properties under idealized conditions, the application of SILMs may require consideration of other practical factors such as support-layer morphology, long-term stability, and scalability.

Conflicts of interest

The authors declare no competing financial interest.

Data availability

Supplementary information is available. See DOI: <https://doi.org/10.1039/d5cp04023j>.

All data included in this study are available upon request by contacting the corresponding author.

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